

Mathew Titus

ClearWater Analytica, LLC

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PERSONAL DATA Citizenship: U.S.A.

EDUCATION

- *Ph.D. in mathematics*, Oregon State University, Oregon, September 2017
Advisor: Edward Waymire
- *B.A. in mathematics and physics*, Willamette University, Oregon, May 2009

INTERESTS

Complex adaptive systems, network evolution,
watershed modeling, harmful algal bloom prediction,
mixing times and cutoff phenomena, fractal fields, and branching processes

ACADEMIC POSITIONS

- Postdoctoral Researcher (DARPA funded)
College of Earth, Ocean, and Atmospheric Sciences, Oregon State University
August 2017 - February 2020
- Research Assistant (NSF funded)
Department of Mathematics, Oregon State University
June 2017 - August 2017
- Graduate Teaching Assistant
Department of Mathematics, Oregon State University
September 2010 - August 2017
- Research Assistant
Department of Electrical Engineering and Computer Science, Oregon State University
June 2014 - September 2014
- Mathematics Instructor
Department of Mathematics, Oregon State University
June 2011 - September 2011; September 2013 - August 2014

ENTREPRENEURSHIP

- CTO, Managing Member
ClearWaterAnalytica, LLC
April 2021 - present

Cloud computing architect and web designer for a Software as a Service (SaaS) product, operationalizing data collection, processing, and machine learning predictions for reservoir management. Partnered with an engineering firm to lead a research effort investigating the utility of eXplainable Artificial Intelligence (XAI) in risk quantification for algal blooms.

- CTO, Managing Member
The Prediction Lab, LLC
June 2018 - present

Worked on fundamental research to create a Bayesian Model Average approach to predicting oncoming harmful algal blooms (HABs) at Detroit Lake, Oregon. Improved client-public relations through speaking engagements and data visualizations. A blog and web application were made to display the state-of-the-reservoir and current risk levels. Managed subcontractors in modeling, satellite data collection, and web design. Also worked as the lead researcher on a DARPA-funded grant investigating new ways to detect and quantify causal connections in complex systems.

- Technical Consultant, Managing Member
Northwest Mathematics, LLC
June 2017 - April 2019

Software design, development and client relations. Created an application and user interface for casino game design, estimating house earnings and jackpot sizes for a variety of parameters and design choices. This involved creating a new probabilistic method of simulating game outcomes that improved on the industry standard for such simulations.

SCHOLARSHIP AND CREATIVE ACTIVITY

Publications

In preparation

- **Mathew Titus**, *Fractal Behavior in Dispersion and Schrödinger Equations*.

Refereed journal publications

- J1. Yevgeniy Kovchegov, Peter T. Otto, and **Mathew Titus**, *Mixing Times for the Mean-Field Blume-Capel Model via Aggregate Path Coupling*, Journal of Statistical Physics, Volume **144**, Number 5 (2011), pp. 1009-1027
- J2. Zachary Gelbaum and **Mathew Titus**, *Simulation of Fractional Brownian Surfaces via Spectral Synthesis on Manifolds*, IEEE Transactions in Image Processing, Volume **23**, Issue 10 (2014), pp. 4383-4388
- J3. James R. Watson, Zachary Gelbaum, **Mathew Titus**, Grant Zoch, and David Wrathall, *Identifying multiscale spatio-temporal patterns in human mobility using manifold learning*, (2020)
- J4. **Mathew Titus** and James R. Watson, *Critical speeding up as an early warning signal for stochastic regime shifts*, (2020)
- J5. **Mathew Titus**, George Hagstrom, and James R. Watson, *Unsupervised manifold learning of collective behavior*, (2021)

Other publications

- O1. **Mathew Titus** , *Mixing Times for Diffusive Lattice-Based Markov Chains*, Thesis.

Professional meetings, symposia, invited talks, and conferences

- 21. April 2022 *Creating a Crystal Ball: The Future of Harmful Algal Bloom Forecasting Using Artificial Intelligence*, AWWA Pacific Northwest Section Conference (**speaker**)
- 20. June 2019 *Predicting Harmful Algal Blooms in Municipal Water Systems*, City Focus Podcast (**speaker**)
- 19. May 2019 *Forecasting Harmful Algal Blooms in Detroit Lake*, North Santiam Basin Summit (**speaker**)
- 18. March 2019 *Predicting Cyanotoxins in Detroit Lake*, City of Salem, OR (**speaker**)
- 17. June 2018 *Unsupervised Detection of Collective Behavior*, Space and Naval Warfare Systems Command (SPAWAR), San Diego, CA (**poster presented**)

16. May 2018 *Unsupervised Detection of Collective Behavior*, OSU Mathematical Biology Seminar (**speaker**)
15. November 2017 *Applications of the Laplace Operator*, Willamette University Mathematics Colloquium (**speaker**)
14. May 2017 *Mixing Times of Diffusive Markov Chains*, OSU Probability Seminar (**speaker**)
13. May 2017 *Southeast Probability Conference*, Durham, NC (**poster presented**)
12. May 2017 *Mixing Times of Diffusive Markov Chains*, University of Oregon Probability Seminar, (**speaker**)
11. May 2017 *Lonseth Lecture Poster Session*, Oregon State University (**presenter**)
10. May 2016 *Frontier Probability Days*, Salt Lake City, UT (**attendee**)
9. January 2016 *Joint Mathematical Meetings of the AMS*, Seattle, WA (**speaker**)
8. May 2015 *SIAM Conference on Applications of Dynamical Systems*, Snowbird, UT (**speaker**)
7. October 2015 *Pacific Northwest Probability Seminar*, Seattle, WA (**attendee**)
6. August 2013 *Mathematical Congress of the Americas*, Guanajuato, MX (**attendee**)
5. July 29 - August 2, 2013 *Summer Short-Course on Mathematics of Climate Change, Related Natural Hazards, and Risks*, CIMAT, Guanajuato, MX (**participant**)
4. June 2012 *Workshop in Geometric Topology*, Oregon State University (**attendee**)
3. April 2012 *Smooth Covering Spaces*, Topology Seminar, Oregon State University (**speaker**)
2. June 2011, *Summer Graduate School in Commutative Algebra*, Mathematical Sciences Research Institute (MSRI), Berkeley, CA (**attendee**)
1. April 2011 *Möbius and Lorentz Transformations*, Mathematics Colloquium Series, Willamette University (**speaker**)

COMMUNITY SERVICE

- Reviewer for American Naturalist.
- Reviewer for Stochastics and Dynamics.
- Reviewer for IEEE Transactions.
- Reviewer for The R Journal.
- Incoming Graduate Student Summer Preparatory Course (Design & instruction), 2016.

COMPUTING LANGUAGES

Python, Matlab, C++, C#, HTML, CSS, Javascript, Bash.

REFERENCES:

- James R. Watson,
james.watson@oregonstate.edu,
College of Earth, Ocean, and Atmospheric Sciences, OSU
- Edward Waymire,
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Department of Mathematics, OSU
- Enrique Thomann,
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Department of Mathematics, OSU